



FACULTY: COMPUTING AND INFORMATION SCIENCE

DEPARTMENT: INFORMATION TECHNOLOGY

MASTERS OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE: ORGANIZATION INFORMATION AND SYSTEM SECURITY

Individual Assignment

The Link:

[https://drive.google.com/drive/folders/1oGEcVDGI2NnajD-Ao2OOsDEquHe_hr6P?usp=drive link](https://drive.google.com/drive/folders/1oGEcVDGI2NnajD-Ao2OOsDEquHe_hr6P?usp=drive_link)

Submitted By:

Submitted To: Dr. NTIVIGURUZZA Jean de la Croix

Reg N°: M04623/2026

Tel:0786230817

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I. STEGOVAULT

I.I User Manual Digital Image Steganography System

1. Introduction

STEGOVAULT is a web-based Digital Image Steganography application that allows users to hide secret messages inside digital images (primarily PNG/JPG) using various Least Significant Bit (LSB) techniques and channel specific methods. It also supports decoding (extracting) the hidden messages.

The system uses:

- **Frontend:** React + Vite + Tailwind CSS
- **Backend:** Node.js/Express + Sharp (image processing) + Multer
- **Core Technique:** Multi-key spatial LSB embedding with optional XOR cipher

Key Features:

- 8 Steganographic Keys (profiles) with different capacities and security levels
- Visual comparison matrix (original vs. stego image)
- Password-protected XOR cipher (K8)
- Real-time preview and download
- Low visible distortion (0.00% in screenshots)

2. System Requirements

Hardware:

- Modern computer with at least 16GB RAM
- Good internet (for dev setup)

Software:

- Node.js (v18+)

- Browser (Chrome/Firefox recommended)

3. Installation & Setup

Backend Setup

Bash

```
cd stego-vault/backend
```

```
npm install
```

- `node server.js`
- Runs on : [SUCCESS] StegoVault Backend Infrastructure Engine Active
- [ROUTING] Listening on: `http://localhost:5174`

Frontend Setup

Bash

```
cd stego-vault/frontend
```

```
npm install
```

```
npm run dev
```

- Runs on <http://localhost:5173>

Note: Ensure CORS allows localhost:5173 to connect to backend 5174.

4. User Interface Overview

The interface is divided into:

1. **Header** : Title + Encode/Decode toggle
2. **Sidebar** : Steganographic Keys (K1–K8)
3. **Main Workspace** : Image upload, message input, controls, output
4. **Footer** : Project info

5. Steganographic Keys (Profiles)

Key ID	Name	Capacity	Security	Target Channels	Description
K1	LSB-1 Classic	3 bits/px	Medium	R, G, B	Standard LSB on RGB
K2	LSB-2 Dual Bit	6 bits/px	Medium	R, G, B	Embeds 2 bits per pixel
K3	LSB-4 Quad Bit	12 bits/px	Low	R, G, B	High capacity, more visible
K4	Red Channel	1 bit/px	High	Red only	Single channel
K5	Green Channel	1 bit/px	High	Green only	Single channel
K6	Blue Channel	1 bit/px	High	Blue only	Single channel
K7	Alpha Channel	1 bit/px	High	Alpha	Uses transparency
K8	XOR Cipher + LSB	3 bits/px	High	R, G, B	LSB + Password XOR

Recommendation:

- **Everyday use: K1** (balanced)
- **Maximum security: K8** (with strong password)
- **Maximum capacity: K3**

6. How to Use Encode Mode

1. **Select Mode:** Click Encode (green button)
2. **Choose Key:** Click a key in the left sidebar (K1 is default)
3. **Upload Cover Image:**
 - Drag & drop or click the upload area
 - Supported: PNG, JPG (PNG recommended for lossless output)
4. **Enter Secret Message:**
 - Type your message in the "SECRET MESSAGE TEXT" box
5. **Optional Password** (for K8):
 - Enter password if using K8

6. **Click "Encode Message into Image"**

7. **Result:**

- Success message
- Download button for the stego image (.png)
- Visual comparison matrix (Original vs Stego)

Screenshots Reference:

- First image: Successful encode with "Hello Dr" message
- Second image: Side-by-side comparison

7. How to Use - Decode Mode

1. **Select Mode:** Click **Decode** (green button)
2. **Choose Matching Key:** Must use the **same key** used during encoding
3. **Upload Stego Image:**
 - Upload the output image from encode step
4. **Enter Password** (if K8 was used)
5. **Click "Extract Hidden Message"**
6. **Result:**
 - Extracted text appears in the right panel
 - Image preview shown

Screenshots Reference:

- Third image: Decode interface showing successful extraction of "Hello Dr"

8. Best Practices

- **Use PNG** as cover and output format (lossless)
- **Keep messages reasonable** (capacity depends on image size)

- **Use strong passwords** for K8
- **Match encoding/decoding keys exactly**
- **Test with small messages** first
- **Avoid compressing** the stego image (JPEG compression destroys LSB data)

9. Technical Details

- **Embedding:** LSB modification on pixel channels
- **Framing:** Messages wrapped with SV START and SV END markers
- **XOR Cipher (K8):** Bitwise XOR with repeating password
- **Image Processing:** Sharp.js for fast pixel manipulation
- **Security:** No visible distortion (confirmed in UI)

Limitations:

- Large messages may exceed image capacity
- K3 (high capacity) may show slight noise
- Password is not stored must remember it

10. Troubleshooting

Common Issues:

- **"No hidden payload":** Wrong key used or image corrupted
- **Upload fails:** Image too large (>50MB)
- **Backend not responding:** Start backend first (node server.js in backend)
- **CORS error:** Check ports and CORS config

Logs:

- Check browser console and backend terminal

11. Project Structure (Key Files)

stego-master/

└─ frontend/ # React UI

| └─ src/

| | └─ components/ # Sidebar, Workspace, Header , Footer, ComparisonView.

| | └─ App.jsx

| └─ package.json

└─ backend/ # Node.js API

| └─ server.js

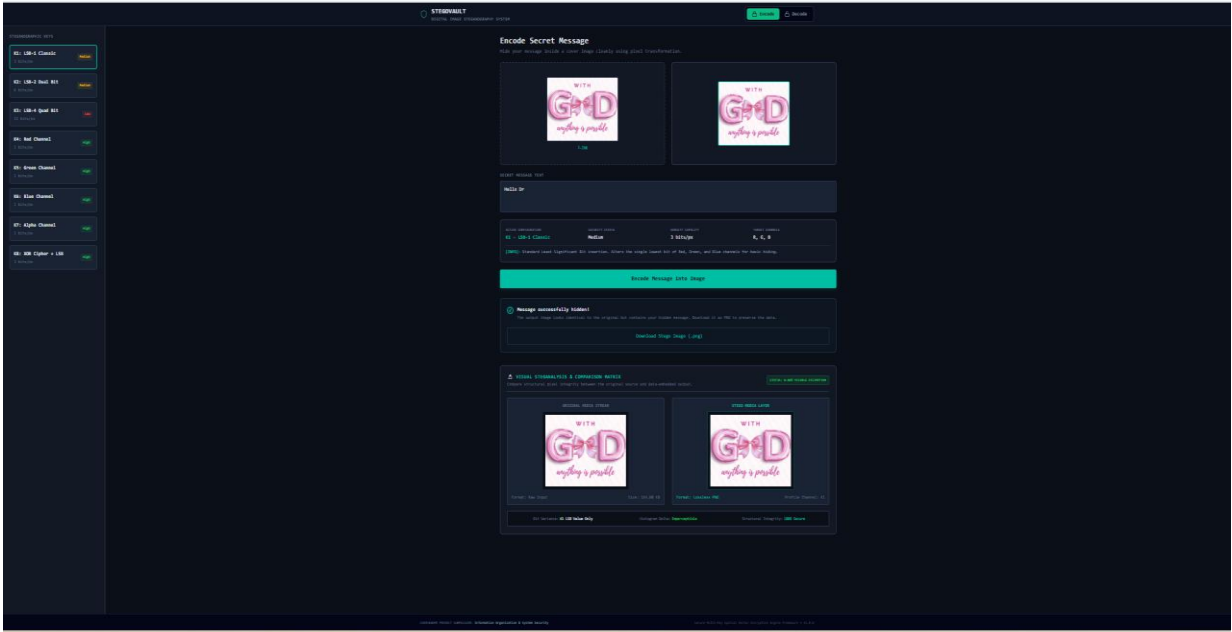
| └─ utils/stegoEngine.js

| └─ package.json

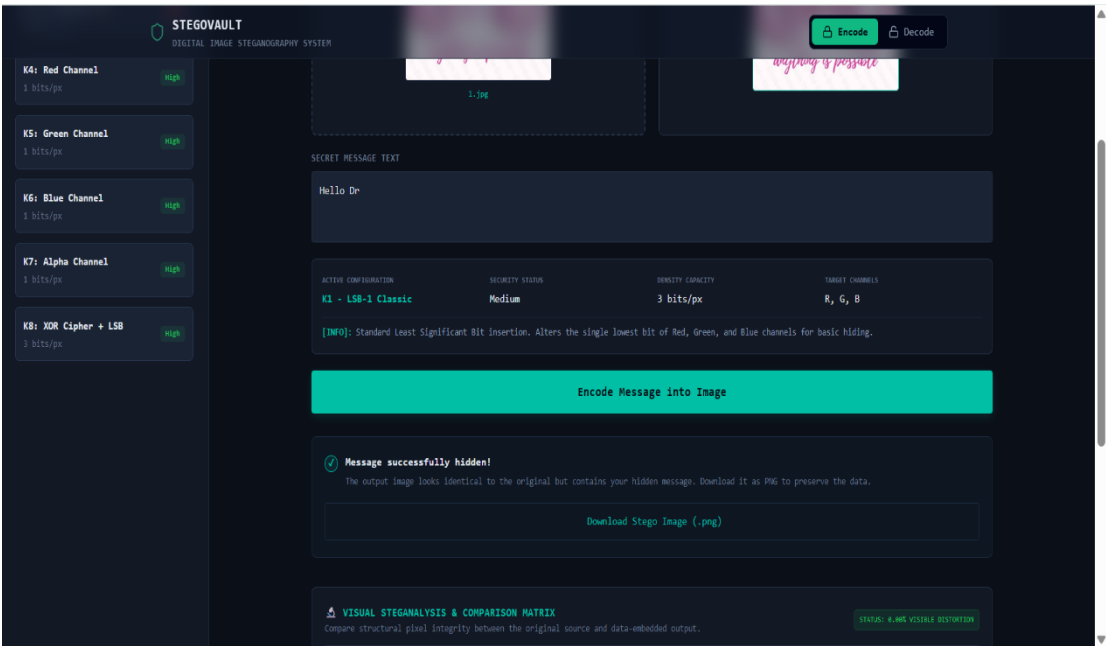
└─ uploads/ # Temporary files

12. Screenshots Guide:

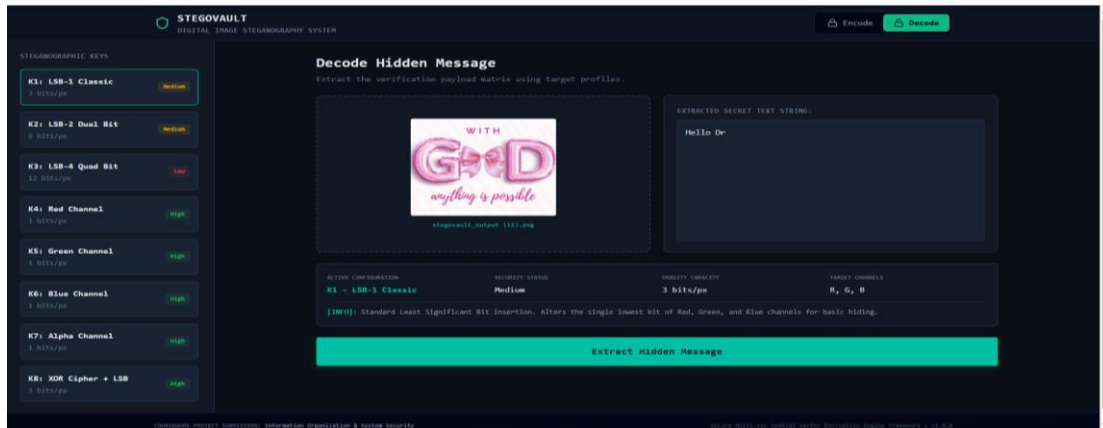
Screenshot 1: Encode success with download option



Screenshot 2: Detailed comparison view (original vs stego)



Screenshot 3: Decode interface with extracted message



The system is designed for educational and demonstration purposes in steganography and information security.

12.Conclusion

StegoVault provides an efficient and secure method for hiding and extracting secret information within digital images. Through multiple embedding techniques and optional password-based encryption, the system enhances data confidentiality while maintaining ease of use.

The link: